

Exam in Statistical Physics and Solid State Physics

Exam date: 17.1.2024

Exam time: 4.5 hours.

Permitted help: books, notes, computer with ITX Flex running, camera for scanning the results. No phones, no internet apart from what is needed for the digital exam.

Answers can be given in English or Danish.

1 Crystal Structure (3 points)

Figure 1 shows a two-dimensional crystal. A couple of distances are given in the drawing. Note that $\delta > 0$.

1. Add the lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ spanning the Bravais lattice to Fig. 1 (i.e., re-draw or copy/paste the lattice to your solution sheet and add the vectors) and draw the smallest possible unit cell.
2. How many atoms does the basis for this lattice contain?
3. In an additional figure, draw the corresponding reciprocal lattice vectors $\mathbf{b}_1$ and $\mathbf{b}_2$. In order to answer this part correctly, the length of $\mathbf{b}_1$ and $\mathbf{b}_2$ needs to be correct, as well as the orientation of $\mathbf{b}_1$ and $\mathbf{b}_2$ relative to each other and to $\mathbf{a}_1$ and $\mathbf{a}_2$.

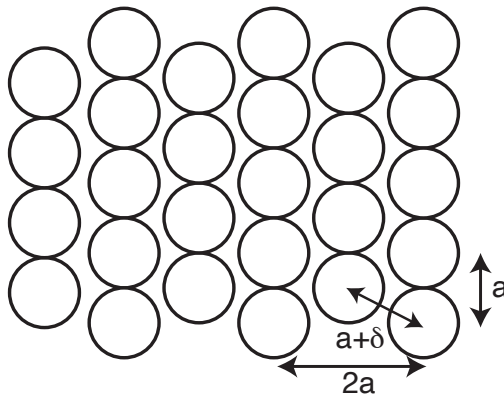


Figure 1: Two-dimensional crystal.

Solution: The vectors are horizontal of length $2a$ and vertical of length a . The unit cell is face-centred rectangular with two atoms in the basis. The reciprocal lattice vectors are horizontal of length π/a and vertical of length $2\pi/a$.

2 The reciprocal lattice (1 point)

Consider a three-dimensional Bravais lattice $\mathbf{R} = m\mathbf{a}_1 + n\mathbf{a}_2 + o\mathbf{a}_3$ and the corresponding reciprocal lattice $\mathbf{G} = m'\mathbf{b}_1 + n'\mathbf{b}_2 + o'\mathbf{b}_3$. For the correct choice of the vectors spanning the reciprocal lattice $\mathbf{b}_1, \mathbf{b}_2$ and $\mathbf{b}_3$, it can be shown that

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij} \quad (1)$$

Making use of this relation, show that

$$e^{i\mathbf{G}\cdot\mathbf{R}} = 1. \quad (2)$$

Solution: $\mathbf{G} \cdot \mathbf{R} = (m\mathbf{a}_1 + n\mathbf{a}_2 + o\mathbf{a}_3) \cdot (m'\mathbf{b}_1 + n'\mathbf{b}_2 + o'\mathbf{b}_3) = 2\pi(mm' + nn' + oo')$ which is an integer multiple of 2π and hence fulfils equ. (2).

3 Speed of sound (1 point)

Figure 2 shows the vibrational modes for a one-dimensional solid. Let the lattice constant $a = 3.14 \text{ \AA}$. Estimate the speed of sound in this solid.

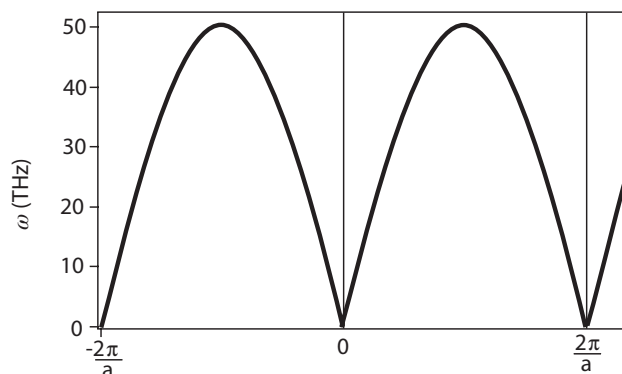


Figure 2: Vibrational modes for a one-dimensional solid.

Solution: We need to know the slope of the linear part of the dispersion to calculate the phase velocity ω/k (or the group velocity, doesn't matter for a linear dispersion). Extrapolating the linear part of the dispersion to the BZ boundary gives $\omega \approx 65 \text{ THz}$ at $k = 2\pi/a \approx 1 \text{ \AA}^{-1}$. So the speed of sound is approximately $65 \text{ THz} / \text{\AA} = 6300 \text{ m/s}$.

4 Heat capacity (1 point)

The Debye temperatures of gold and beryllium are 162 and 1440 K, respectively. Which material has the higher molar heat capacity at 77 K and why?

Solution: Gold has the higher molar heat capacity because the lower Debye temperature means that more oscillators are “unfrozen” or, more formally, because $C = \frac{12\pi^4}{5} Nk_B \left(\frac{T}{\Theta_D} \right)^3$ which is a low temperature limit but still ok here for Au.

5 Electronic states in crystals (1 point)

The wave function of an electronic state in a one-dimensional crystal with lattice constant a is a Bloch wave

$$\psi_k(x) = e^{ikx} u_k(x). \quad (3)$$

Consider a point x' on the lattice and a wave vector $k = \pi/a$. Evaluate the Bloch wave at the position $x'' = x' + 2a$. Write the result as a function of $\psi_k(x')$, the value at x' .

Solution: $\psi_k(x'') = \psi_k(x')$ because $u_k(x)$ is lattice periodic and $e^{ik(x'+2a)} = e^{ikx'}$ for $k = \pi/a$.

6 Metals and Insulators (2 points)

Beryllium is a group II element with two valence electrons per atom. It crystallises in the hexagonal closed packed crystal structure which has two atoms per unit cell. Using this information, can you infer if Be is a metal or an insulator? Explain your reasoning.

Solution: Since one Be atom has two electrons per atom and the hcp structure has two atoms per unit cell, there are four electrons per unit cell. For an even number of valence electrons per unit cell, we can't say if the material is an insulator or a metal. In the present case, four valence electrons could fill two bands and turn the material into an insulator. Or they could fill more than two band partially, turning it into a metal (this is actually the case).

7 Density of states (3 points)

Fig. 3 shows the band structure near the band gap of a semiconductor. Make a qualitative sketch of the density of states in the same energy region.

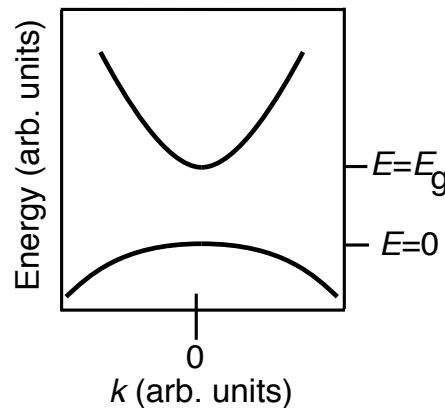


Figure 3: Band structure of a semiconductor near the band gap.

Solution: The required density of states is given by $g_C(E) = \frac{V}{2\pi^2} \left(\frac{2m_e^*}{\hbar^2} \right)^{3/2} (E - E_g)^{1/2}$ and $g_V(E) = \frac{V}{2\pi^2} \left(\frac{2m_h^*}{\hbar^2} \right)^{3/2} (-E)^{1/2}$ for the conduction band and valence band, respectively. Both are functions of the form $\sqrt{E}$, one starting at zero and one starting at E_g and both increasing away from the energy gap. 1 p for a drawing like this, 1 p for $g_V(E)$ rising more steeply because of the higher effective mass seen in the figure.

8 Band structure (3 points)

Figure 4 shows the band structure of two semiconductors along high-symmetry directions in the Brillouin zone. Assume that the structure and lattice constant is the same for both materials, so that the horizontal axis and the distances in reciprocal space are also the same. The energy zero is chosen to match the top of the valence bands.

1. Discuss if the materials have a direct or an indirect band gap.

- Judging from the dispersion in the figure, which material has the lower effective mass near the conduction band minimum?

Give brief reasons for your answers.

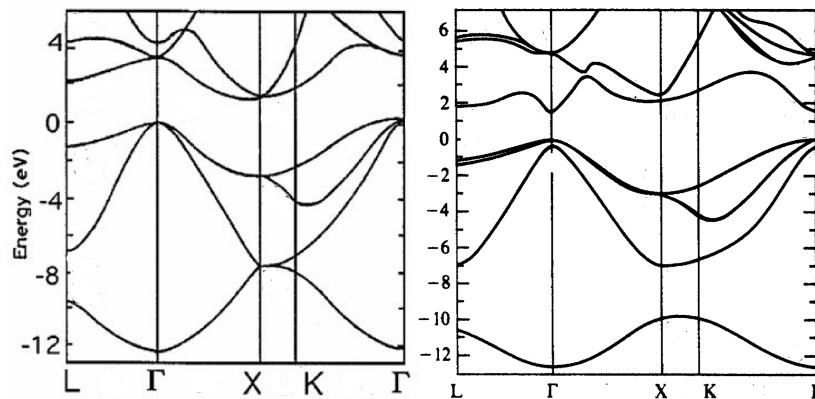


Figure 4: Band structures of two semiconductors.

Solution: The left material has an indirect band gap because the CBM is not at the same k as the VBM. For the right hand material it is and thus this is a direct band gap semiconductor. The right hand material has the lower effective mass because the band near the CBM is clearly curved more strongly.

9 Dielectrics (1 point)

The dielectric constant of air is just slightly higher than 1. It is 1.000573. What is the main reason for it being so much smaller than the dielectric constant of silicon that is 11.7?

Solution: The density in any solid is much higher than in air, so there are more atoms to be polarised (plus there are also additional polarisation mechanisms.)

10 Magnetic ordering (2 points)

Consider an elemental ferromagnetic solid that is well-described by the Heisenberg model of spin $1/2$ magnetic moments on a lattice. What is the highest possible magnetisation

- per atom.
- per unit cell when the magnetic material crystallises in the face-centred cubic structure.

Solution: We take the gyromagnetic ratio for the electron to be two and so the highest possible magnetisation per atom is μ_B . This is also the highest magnetisation per unit cell because the fcc structure is primitive.

11 Partition function

1) The partition function is

$$Z = \sum_{i=1}^4 n_i e^{-\epsilon_i \beta}, \quad (1)$$

where $\beta = 1/(kT)$, and k is Boltzmann's constant.

2) We obtain the mean energy as

$$\bar{E} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = \frac{n_1 \epsilon_1 e^{-\epsilon_1 \beta} + n_2 \epsilon_2 e^{-\epsilon_2 \beta} + n_3 \epsilon_3 e^{-\epsilon_3 \beta} + n_4 \epsilon_4 e^{-\epsilon_4 \beta}}{n_1 e^{-\epsilon_1 \beta} + n_2 e^{-\epsilon_2 \beta} + n_3 e^{-\epsilon_3 \beta} + n_4 e^{-\epsilon_4 \beta}}. \quad (2)$$

The limit $T \rightarrow \infty$ corresponds to $\beta \rightarrow 0$. Inserting $\beta = 0$ in the expression for $\bar{E}$, we get

$$\lim_{T \rightarrow \infty} \bar{E} = \frac{n_1 \epsilon_1 + n_2 \epsilon_2 + n_3 \epsilon_3 + n_4 \epsilon_4}{n_1 + n_2 + n_3 + n_4}. \quad (3)$$

Physically, we obtain this limit, because all microstates are equally probable at high temperature, as this maximizes the entropy.

The limit $T \rightarrow 0$ corresponds to $\beta \rightarrow \infty$. To determine this limit we rewrite the expression for the mean energy into

$$\bar{E} = \frac{n_1 \epsilon_1 + n_2 \epsilon_2 e^{-(\epsilon_2 - \epsilon_1) \beta} + n_3 \epsilon_3 e^{-(\epsilon_3 - \epsilon_1) \beta} + n_4 \epsilon_4 e^{-(\epsilon_4 - \epsilon_1) \beta}}{n_1 + n_2 e^{-(\epsilon_2 - \epsilon_1) \beta} + n_3 e^{-(\epsilon_3 - \epsilon_1) \beta} + n_4 e^{-(\epsilon_4 - \epsilon_1) \beta}}. \quad (4)$$

For $\beta \rightarrow \infty$, all the exponentials approach zero. We are hence left with

$$\lim_{T \rightarrow 0} \bar{E} = \epsilon_1. \quad (5)$$

Physically, the limit takes this value, because at low temperature, the system is in the ground state.

12 Photons in a kiln

From (7.72) in Schroeder, the average number of photons in the mode is

$$\bar{n} = \frac{1}{e^{hf/(kT)} - 1}. \quad (6)$$

When $kT \gg hf$, this can be approximated as

$$\bar{n} \approx \frac{kT}{hf}. \quad (7)$$

So

$$a = \frac{k}{hf}. \quad (8)$$

13 Two-state paramagnets

1) The net flow of energy is from 2 to 1. This is so, because the entropy of 2 increases more when 2 gives away a bit of energy than the entropy of 1 would increase if 1 gave away the same amount of energy. The entropy is hence maximized by 2 giving energy to 1 until they have reached the same temperature.

2) The temperatures fulfil $T_1 < T < T_2 < 0$. We saw in the previous question that the entropy is maximized by 2 giving energy to 1 until they have reached the same temperature. The temperature of paramagnet 2 becomes more negative when 2 gives away energy to reach thermal equilibrium, while the temperature of paramagnet 1 becomes less negative when 1 gains energy to reach thermal equilibrium. Hence $T_1 < T < T_2 < 0$.

14 Degenerate Fermi gas

From (7.39) in Schroeder, we have

$$\bar{n}_{\text{DFG}} = \frac{\pi(8m\mu)^{3/2}}{3h^3} \quad \text{for } \tilde{\mu} = 0 \text{ and } \mu \geq 0. \quad (9)$$

Half of this density is spin up particles and the other half is spin down particles.

Let us now consider $\tilde{\mu} \neq 0$. For spin up particles, states with kinetic energies up to $\mu + \tilde{\mu}B$ are filled, and for spin down particles states with kinetic energies up to $\mu - \tilde{\mu}B$ are filled. The above expression is hence modified to

$$\bar{n}_{\text{DFG}} = \frac{\pi(8m(\mu + \tilde{\mu}B))^{3/2}}{6h^3} + \frac{\pi(8m(\mu - \tilde{\mu}B))^{3/2}}{6h^3} \quad \text{for } \mu > |\tilde{\mu}B|. \quad (10)$$